MEDIUM

ADVANCED MATH

Advanced Math

12

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

If $x \neq 0$, which of the following expressions is equivalent to $\frac{\sqrt{16x^4y^8}}{x^3}$?

A. $8x^2y^4$

B. $4xy^4$

C. $4x^{-2}y^2$

D. $4x^{-1}y^4$

Q2 ADVANCED MATH

The product of two positive integers is 546. If the first integer is 11 greater than twice the second integer, what is the smaller of the two integers?

A. 7

B. 14

C. 39

D. 78

Q3 ADVANCED MATH

$$(x+2)(x+3)=(x-2)(x-3)+10$$

Which of the following is a solution to the given equation?

A. 1

B. 0

C. -2

D. -5

Q4 ADVANCED MATH

$$y = 0.25x^2 - 7.5x + 90.25$$

The equation gives the estimated stock price y, in dollars, for a certain company x days after a new product launched, where $0 \leq x \leq 20$. Which statement is the best interpretation of $x, y = 1, 83$ in this context?

A. The company's estimated stock price increased $\$\,83$ every day after the new product launched.

B. The company's estimated stock price increased $\$\,1$ every 83 days after the new product launched.

C. 1 day after the new product launched, the company's estimated stock price is $\$\,83$.

D. 83 days after the new product launched, the company's estimated stock price is $\$\,1$.

Q5 ADVANCED MATH

$$p = 20 + \frac{16}{n}$$

The given equation relates the numbers p and n, where n is not equal to 0 and $p > 20$. Which equation correctly expresses n in terms of p?

A. $n = \frac{p - 20}{16}$

B. $n = \frac{p}{16} + 20$

C. $n = \frac{p}{16} - 20$

D. $n = \frac{16}{p - 20}$

Q6 ADVANCED MATH

x	24	30	32
fx	-8	-8	8

For the quadratic function f, the table shows three values of x and their corresponding values of fx. Which equation defines f?

A. $f(x) = (x - 24)(x - 30) + 4$

B. $f(x) = (x - 24)(x - 30) - 8$

C. $f(x) = (x - 8)(x - 32) + 32$

D. $f(x) = (x - 8)(x - 32) - 32$

Q7

GRID-IN ADVANCED MATH

$$x^2 - ax + 12 = 0$$

In the equation above, a is a constant and $a > 0$. If the equation has two integer solutions, what is a possible value of a ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8 GRID-IN ADVANCED MATH

The population of the town of Smithville doubled every 75 years from 1659 to 1959. The population of this town was 240,000 in 1959. What was the population of this town in 1659?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN ADVANCED MATH

$$\frac{-54}{w} = 6$$

What is the solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10 ADVANCED MATH

$$fx = \frac{a - 19}{x} + 5$$

In the given function f, a is a constant. The graph of function f in the xy-plane, where $y = fx$, is translated 3 units down and 4 units to the right to produce the graph of $y = gx$. Which equation defines function g?

A. $gx = \frac{a - 19}{x + 4} + 2$

B. $gx = \frac{a - 19}{x - 4} + 2$

C. $gx = \frac{a - 22}{x + 4} + 5$

D. $gx = \frac{a - 22}{x - 4} + 5$